SIMATS ENGINEERING

**NAME : MURALI KRISHNAN N**

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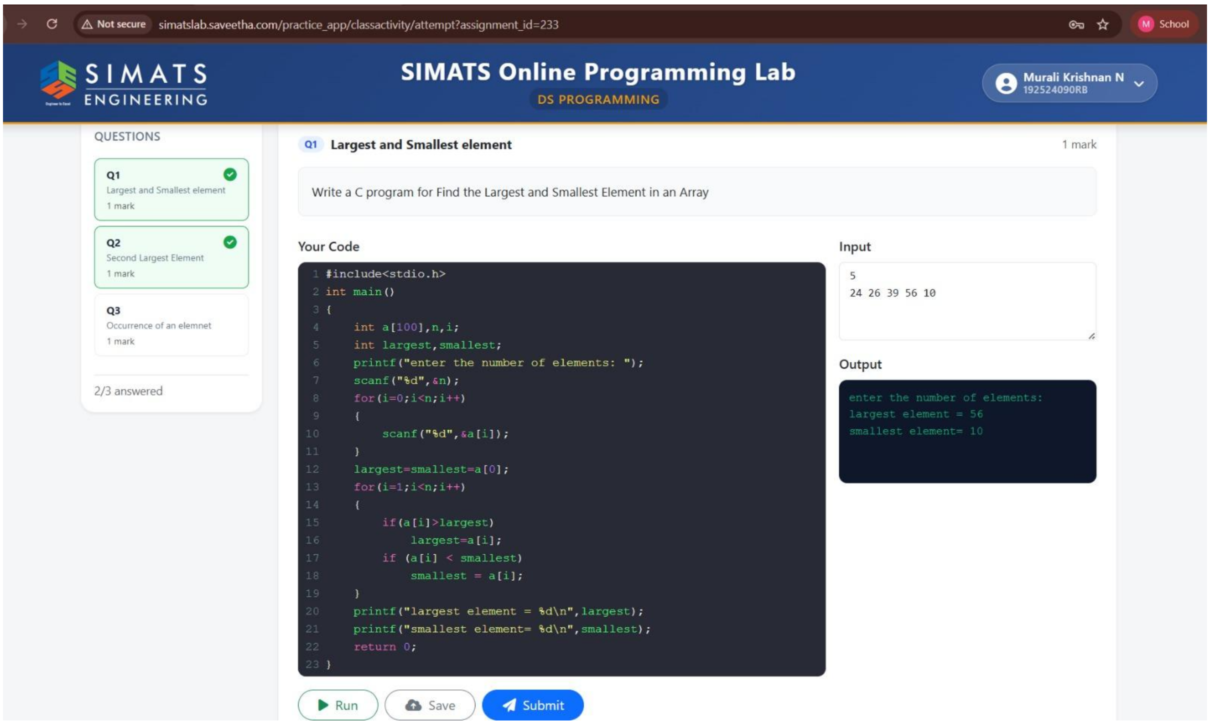
**COURSE CODE: CSA0311**

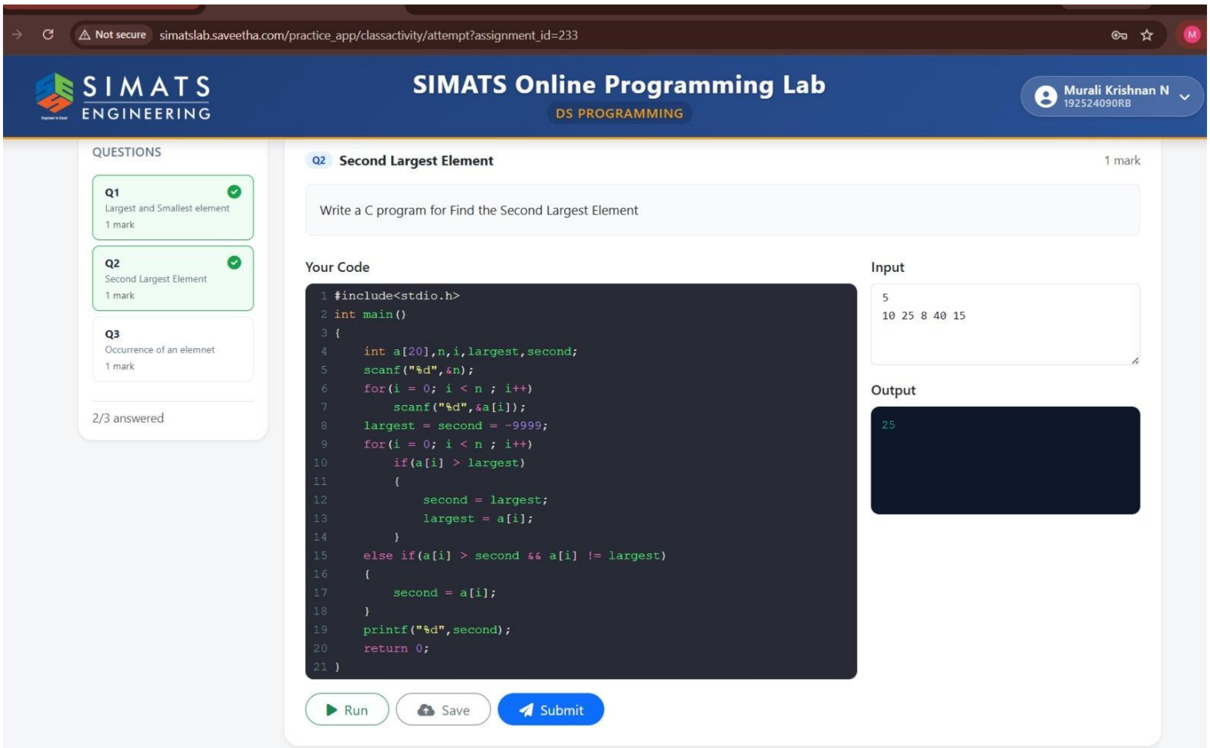
**COURSE NAME: DATA STRUCTURES**

# Output Prediction

# 1.



# 2.



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DS PROGRAMMING

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QUESTIONS

Q1
Largest and Smallest element
1 mark

Q2
Second Largest Element
1 mark

Q3
Occurrence of an elemnet
1 mark

2/3 answered

Q2 **Second Largest Element** 1 mark

Write a C program for Find the Second Largest Element

Your Code

```c
#include<stdio.h>
int main()
{
    int a[20],n,i,largest,second;
    scanf("%d",&n);
    for(i = 0; i < n ; i++)
        scanf("%d",&a[i]);
    largest = second = -9999;
    for(i = 0; i < n ; i++)
        if(a[i] > largest)
        {
            second = largest;
            largest = a[i];
        }
    else if(a[i] > second && a[i] != largest)
    {
        second = a[i];
    }
    printf("%d",second);
    return 0;
}
```

Input

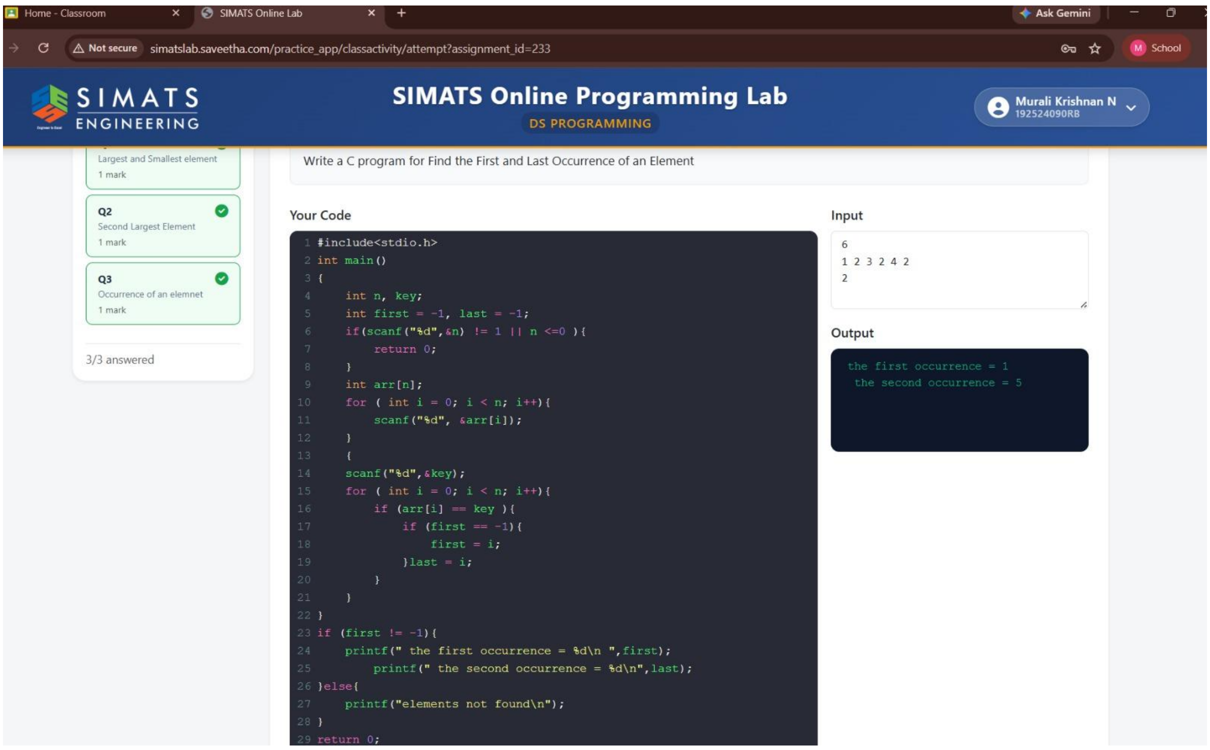
```
5
10 25 8 40 15
```

Output

```
25
```

Run Save Submit

# 3.



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Largest and Smallest element
1 mark

Q2
Second Largest Element
1 mark

Q3
Occurrence of an element
1 mark

3/3 answered

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```c
#include<stdio.h>
int main()
{
    int n, key;
    int first = -1, last = -1;
    if(scanf("%d",&n) != 1 || n <=0 ){
        return 0;
    }
    int arr[n];
    for ( int i = 0; i < n; i++){
        scanf("%d", &arr[i]);
    }
    {
    scanf("%d",&key);
    for ( int i = 0; i < n; i++){
        if (arr[i] == key ){
            if (first == -1){
                first = i;
            }last = i;
        }
    }
}
if (first != -1){
    printf(" the first occurrence = %d\n ",first);
        printf(" the second occurrence = %d\n",last);
}else{
    printf("elements not found\n");
}
return 0;
```

Input

```
6
1 2 3 2 4 2
2
```

Output

```
the first occurrence = 1
 the second occurrence = 5
```